The documentation and process conversion measures necessary to comply with this amendment shall be completed by 21 October 2000.

INCH-POUND

MIL-PRF-19500/583A
AMENDMENT 2
21 July 2000
SUPERSEDING
AMENDMENT 1
29 November 1999

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON AMPLIFIER,
TYPES 2N5681 AND 2N5682 JAN, JANTX AND JANTXV

This amendment forms a part of MIL-PRF-19500/583A, dated 21 August 1999, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* 1.3 notes, second note 1/; delete "1/" and substitute "2/".

PAGE 6

* 4.4.1; delete and substitute the following:

"4.4.1 Group A inspection shall be conducted in accordance with table I herein."

PAGE 7

* 4.4.2; delete and substitute as follows:

" 4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified. Separate samples may be used for each step. In the event of a group B failure, the manufacturer may pull a new sample at double the sample size from either the failed assembly lot or from another assembly lot from the same wafer lot. If the new "assembly lot" option is exercised, the failed assembly lot shall be scrapped. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein. Delta measurements shall be in accordance with table II herein.

Step	Method	Condition
1	1027	Steady-state life: 340 hours, $V_{CB} \geq 20$ V dc, $T_J = 150°C$ minimum. External heating of the device under test to achieve $T_J = 150°C$ minimum is allowed provided that a minimum of 75 percent of rated power is dissipated. No heat sink or forced-air cooling on the devices shall be permitted. n = 45 devices, c = 0
2	1026	The steady state life test of step 1 shall be extended to 1,000 hours for each die design. Samples shall be selected from a wafer lot every twelve months of wafer production, however, Group B shall not be required more than once for any single wafer lot. n = 45, c = 0.
3	1032	High-temperature life (non-operating), $T_A = +200°C$. n = 22, c = 0. "

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PAGE 7 continued

4.4.2.1, first line; delete “from” and substitute “for”.

* 4.4.2.1a, second line; delete “See MIL-PRF-19500.”

* 4.4.3.1 delete and substitute as follows:

“4.4.3.1 Group C inspection, table VII (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E.
C6		Not applicable.”

* 4.4.4.1, E2, sampling column; delete “32” and substitute “22”.

4.4.4.1, E4, method column; delete “3161” and substitute “3131”.

PAGE 8

4.5.2e, delete “1200” and substitute “10”.

PAGE 11

* Table I, subgroup 5, inspection column; delete “5/”.

Table I, delete note 5/ in its entirety.

PAGE 12

Table II, note 2/; delete and substitute as follows:

“2/ The delta measurements for group B, 4.4.2 are as follows:

a. Step 1, see table II herein, steps 1 and 2.
b. Step 2, see table II herein, steps 1 and 2.”

* Table II, delete note 3/ in its entirety.

* Table II, delete note 4/ and substitute the following as new note 3/ :

“3/ The delta measurements for table IX of MIL-PRF-19500 are as follows:

a. Subgroup E2, see table II, steps 1 and 2.”

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Preparing activity:
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